



JKBOSE 2026

CLASS 11TH

INTRODUCTION TO THREE DIMENSIONAL GEOMETRY

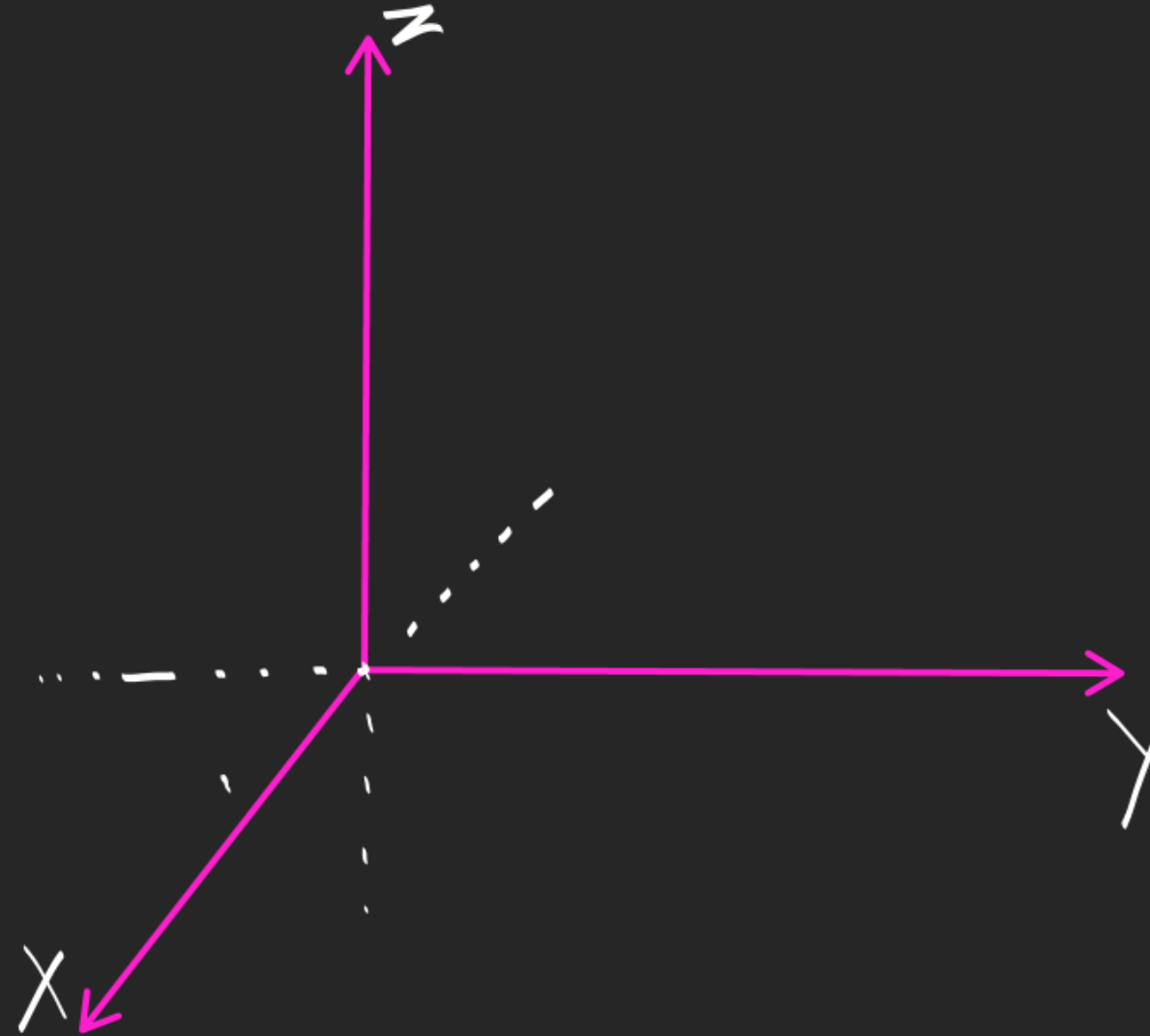
MATHS

LECTURE-1

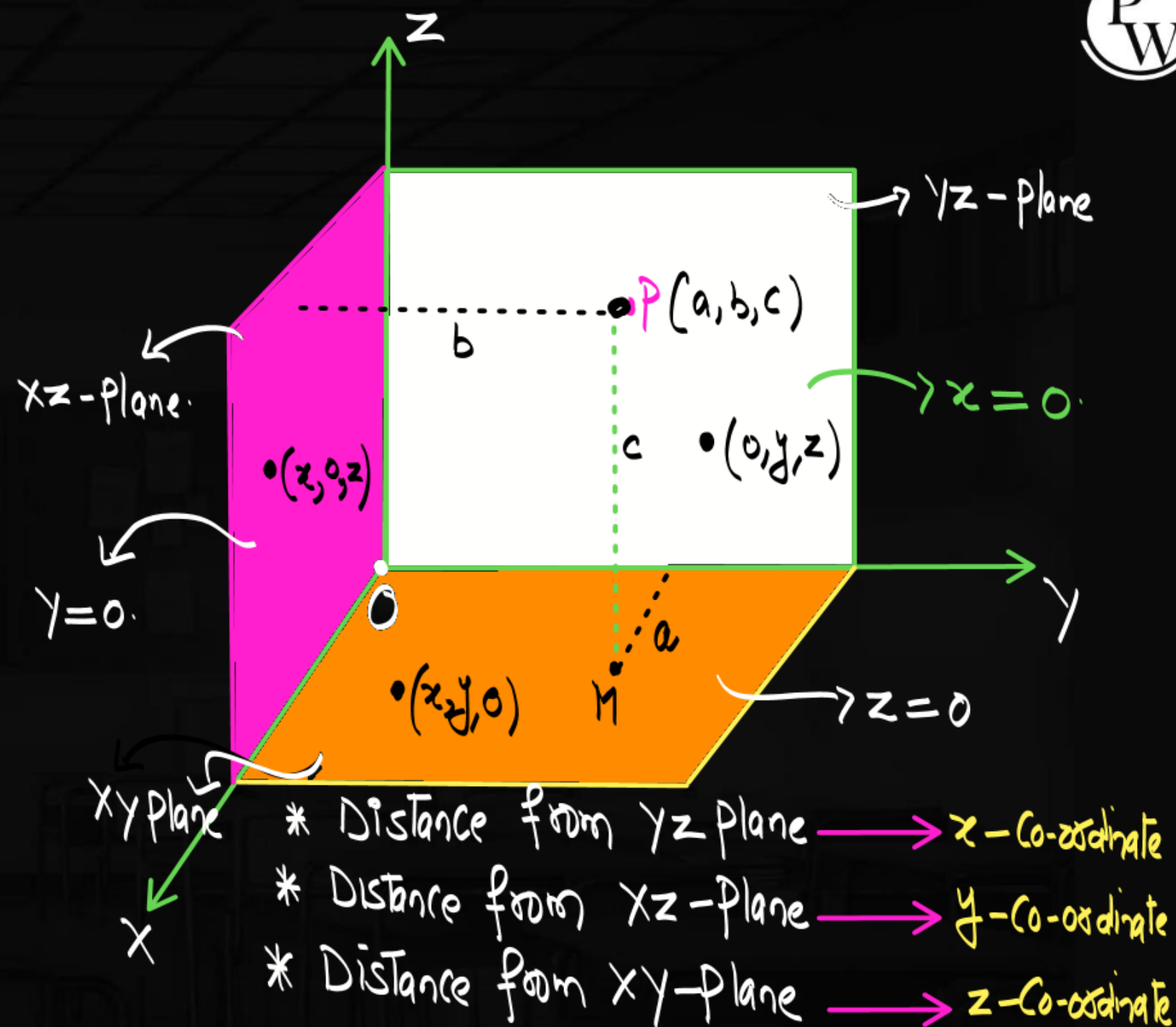
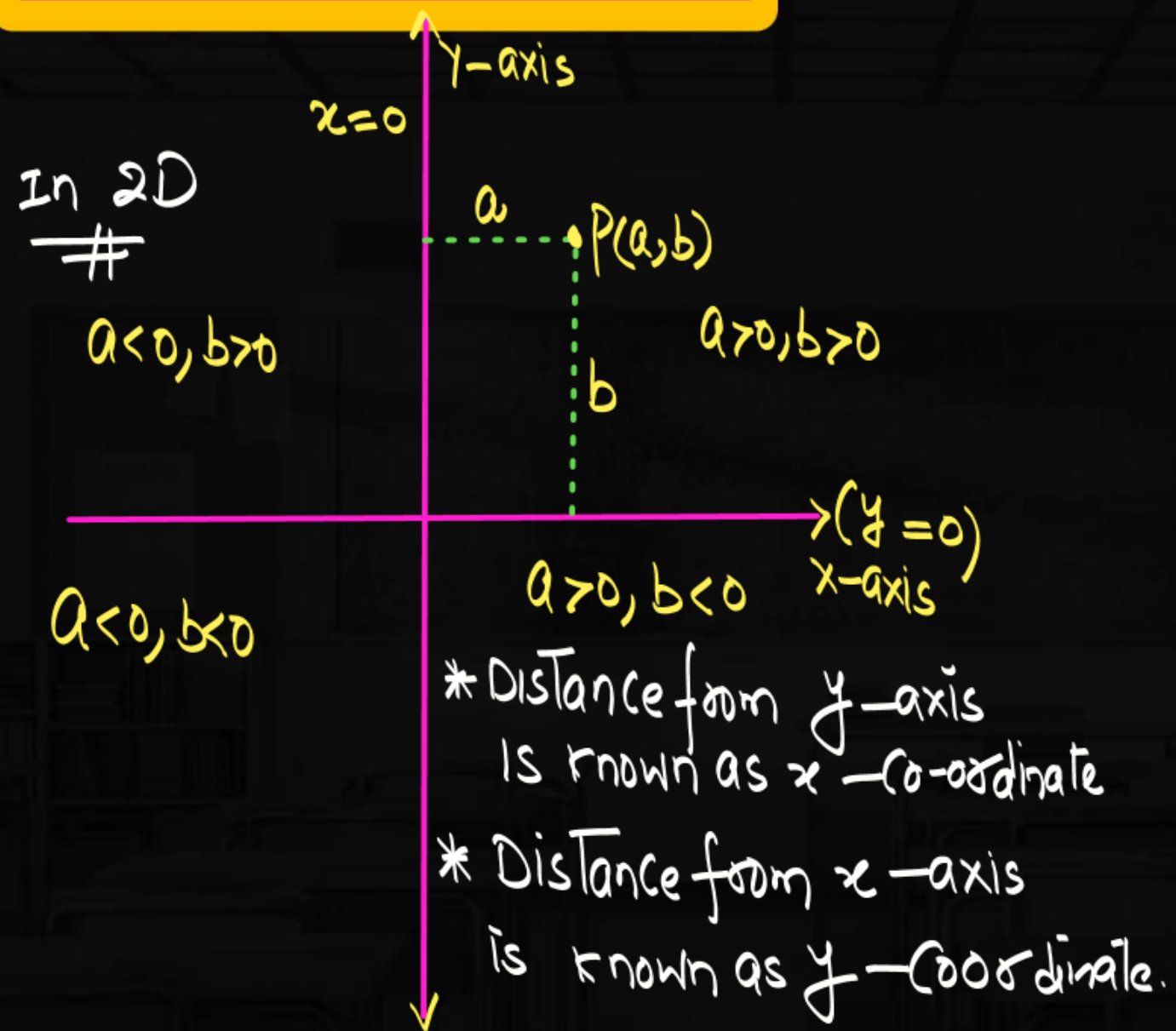
BY-MEHRAJ SIR



TOPICS TO BE COVERED

1**Introduction to Coordinate System****2****Exercise Discussion 11.1****3****Distance Formula****4****Related Problems**

INTRODUCTION



Here * $x=0$ represents yz -Plane.

* $y=0$ represents xz -Plane

* $z=0$ represents xy -Plane.

* $x=0, y=0$ represents z -axis.

* $x=0, z=0$ represents y -axis.

* $y=0, z=0$ represents x -axis.

* The xy , yz & xz Planes divides the space into 8 parts & each part is known as an octant.

Co-ordinates/octants	I	II	III	IV	V	VI	VII	VIII
x	+	-	-	+	+	-	-	+
y	+	+	-	-	+	+	-	-
z	+	+	+	+	-	-	-	-

Octants Coordinate	I	II	III	IV	V	VI	VII	VIII
x	+	−	−	+	+	−	−	+
y	+	+	−	−	+	+	−	−
z	+	+	+	+	−	−	−	−

QUESTION (NCERT EXAMPLER):

The distance of point $P(3, 4, 5)$ from the yz -plane is

1

3 units ✓

2

4 units ✗

3

5 units ✗

4

550 ✗

(Sol) Distance from yz -Plane = x -Co-ordinate.
 $= 3$ units.

QUESTION (NCERT EXAMPLER):

The point $(-2, -3, -4)$ lies in the

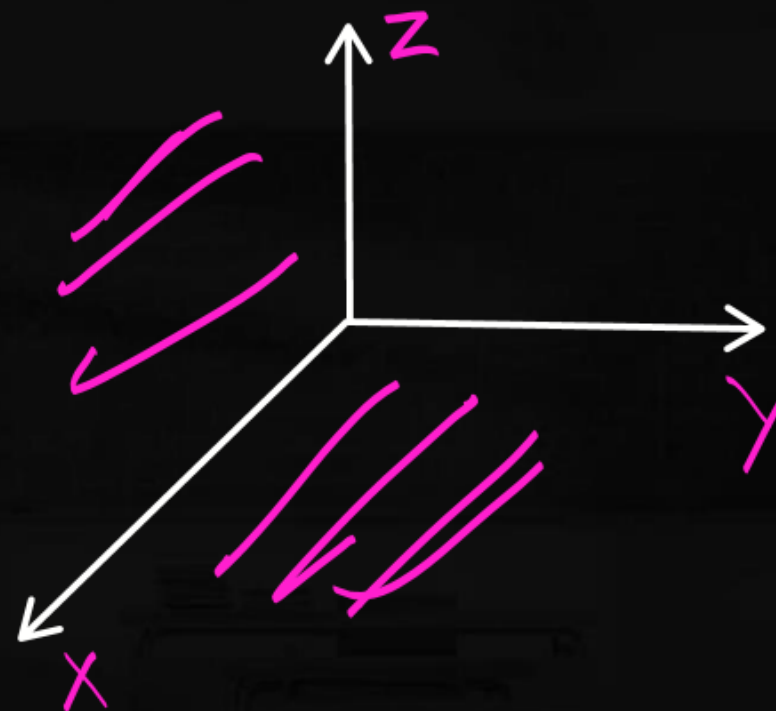
- 1 First octant ✗
- 2 Seventh octant ✓
- 3 Second octant ✗
- 4 Eighth octant ✗

+	-	-	+	+	-	-	+
+	+	-	-	+	+	-	-
+	+	+	+	-	-	-	-

QUESTION (NCERT EXAMPLER):

x -axis is the intersection of two planes

- 1 ☒ xy and xz
- 2 ☒ zy and zx
- 3 ☒ xy and yz
- 4 ☒ None of these



QUESTION (NCERT EXAMPLER):

Equation of y-axis is considered as

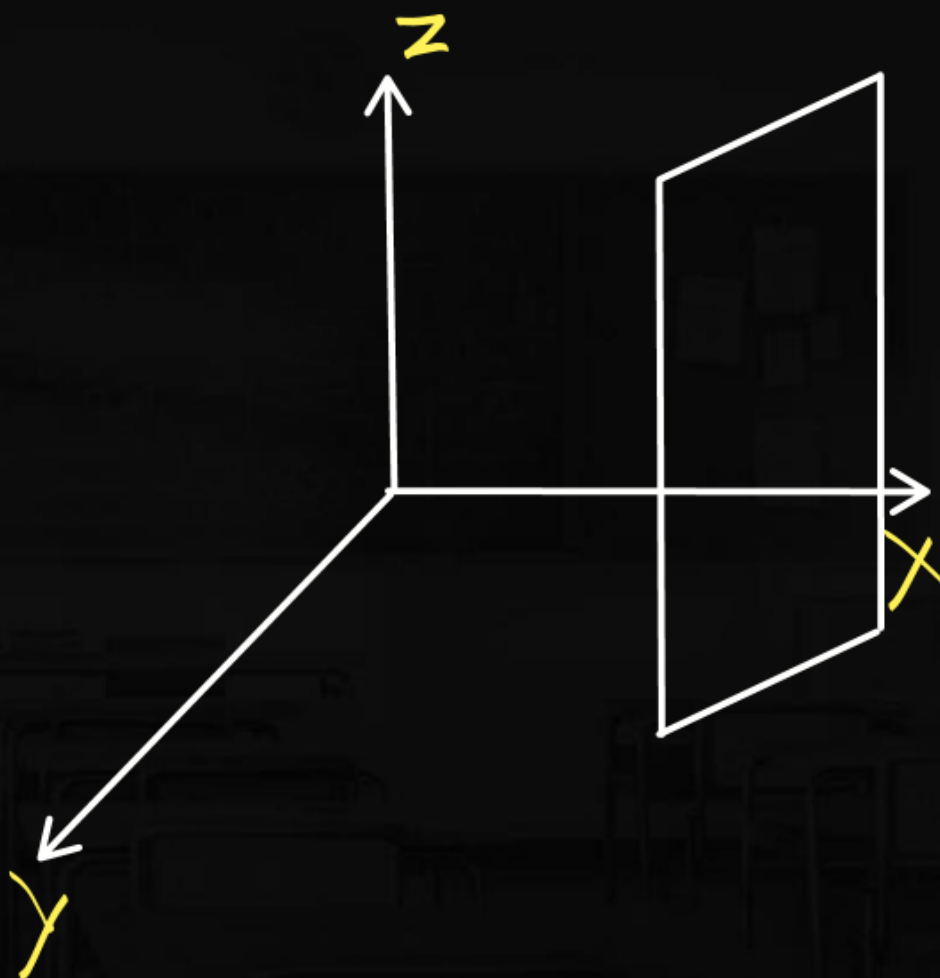
$$x = 0, z = 0$$

- 1** $x = 0, y = 0$ ✗
- 2** $y = 0, z = 0$ ✗
- 3** $z = 0, x = 0$ ✓✓
- 4** None of these

QUESTION (NCERT EXAMPLER):

A plane is parallel to yz-plane so it is perpendicular to:

- 1** x -axis ✓
- 2** y -axis ✗
- 3** z -axis ✗
- 4** None of these ✗



QUESTION (NCERT EXAMPLER):

The locus of a point for which $y = 0, z = 0$ is

1 Equation of x -axis ✓

2 Equation of y -axis ✗

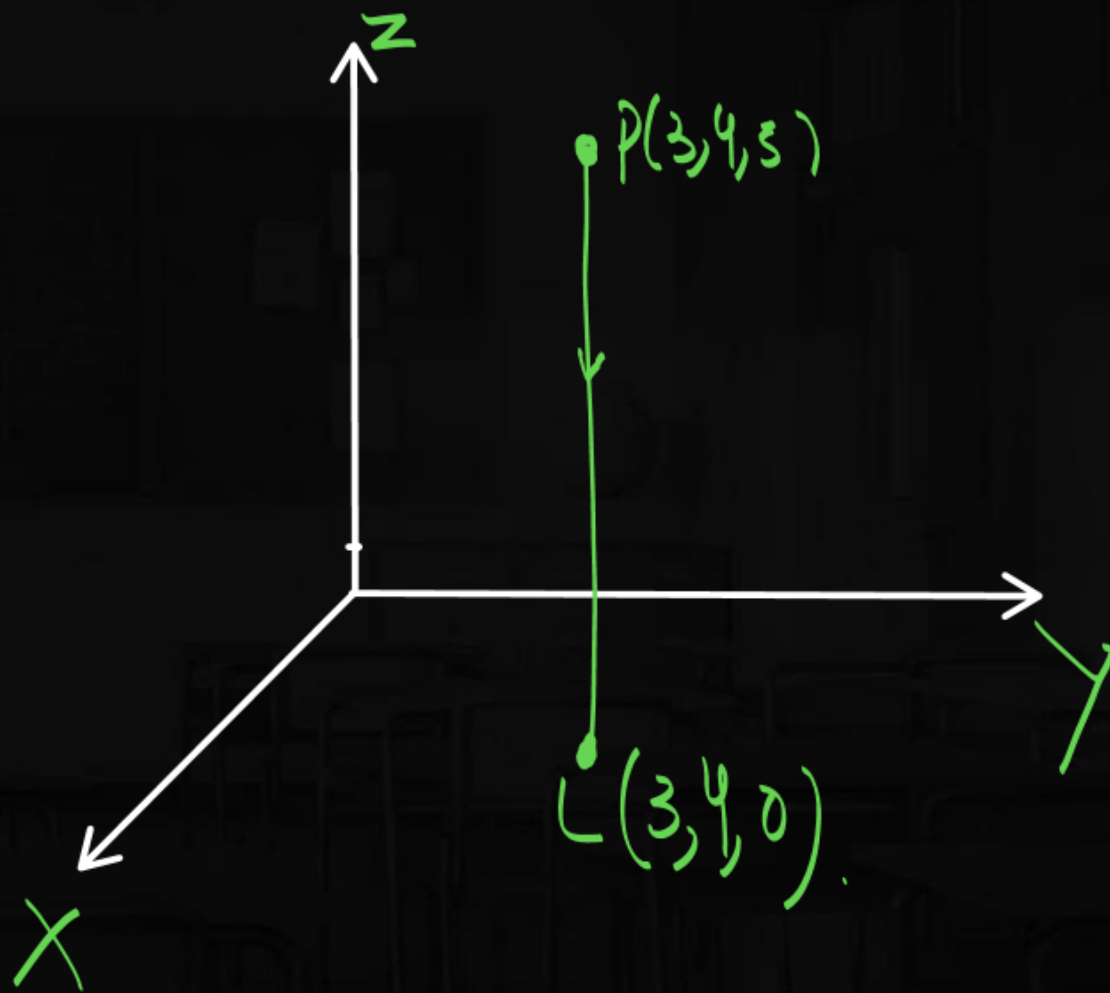
3 Equation z -axis ✗

4 None of these ✗

QUESTION (NCERT EXAMPLER):

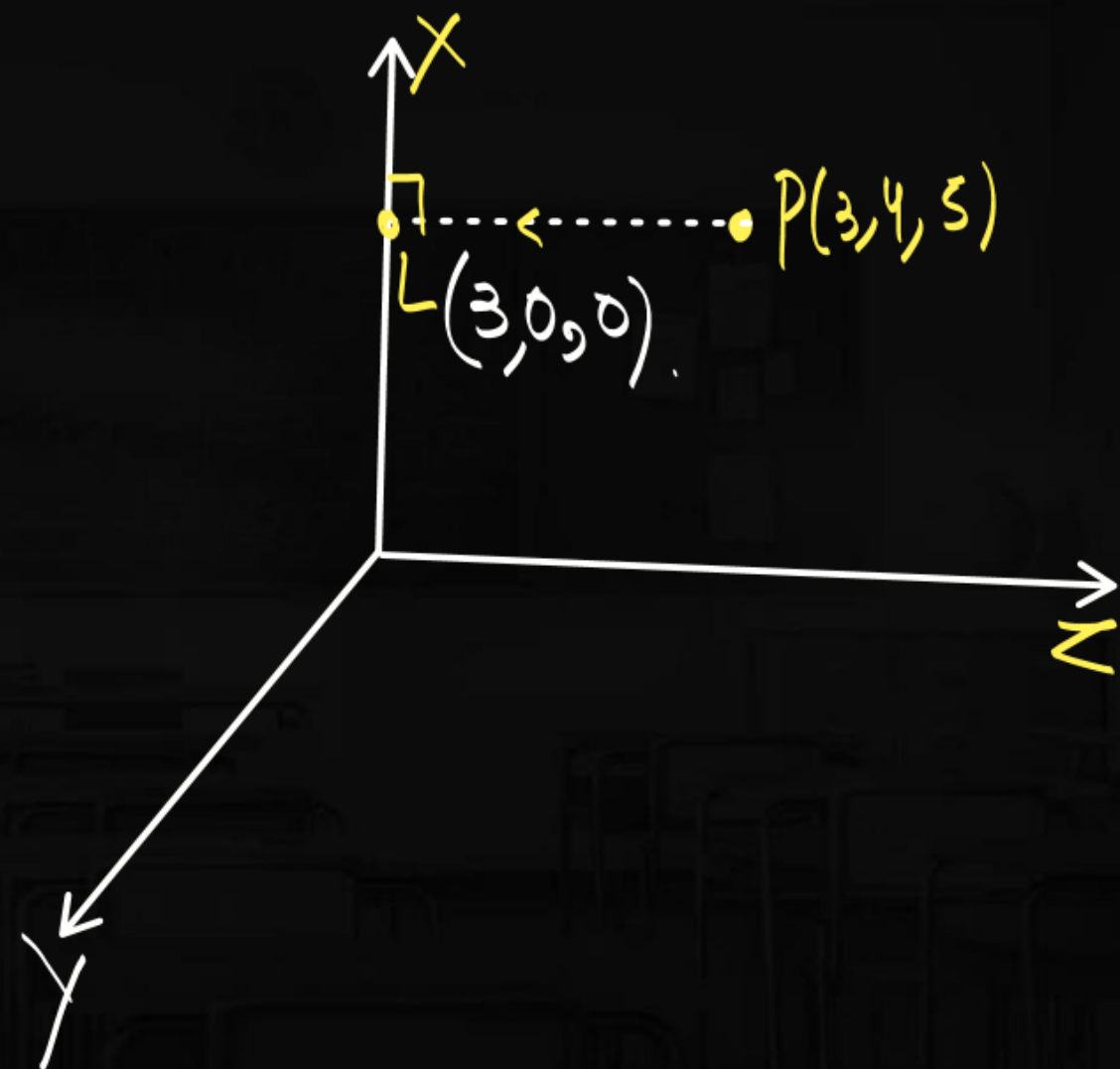
L is the foot of the perpendicular drawn from a point $P(3, 4, 5)$ on the xy -plane.
The coordinates of point L are

- 1** $(3, 0, 0)$ ✗
- 2** $(0, 4, 5)$ ✗
- 3** $(3, 0, 5)$ ✗
- 4** None of these ✓✓



QUESTION (NCERT EXAMPLER):

L is the foot of the perpendicular drawn from a point $(3, 4, 5)$ on x -axis.
The coordinate of L are



- 1** $(3, 0, 0)$ ✓
- 2** $(0, 4, 0)$ ✗
- 3** $(0, 0, 5)$ ✗
- 4** None of these ✗

EXERCISE – 11.1

QUESTION

A point is on the x -axis. What are its y -coordinate and z -coordinates?

(Sol) y -Coordinate $= 0$

z -Coordinate $= 0$.

QUESTION

A point is in the XZ-plane. What can you say about its y -coordinate?

y -Coordinate = 0.

QUESTION

Name the octants in which the following points lie:

$(1, 2, 3), (4, -2, 3), (4, -2, -5), (4, 2, -5), (-4, 2, -5), (-4, 2, 5), (-3, -1, 6), (-2, -4, -7).$

h/w

x	+	-	-	-	+	-	-	+
y	+	+	-	-	+	+	-	+
z	+	+	+	+	-	-	-	-

QUESTION

Fill in the blanks:

- (i) The x -axis and y -axis taken together determine a plane known as XY-plane.
- (ii) The coordinates of points in the XY-plane are of the form $(a, b, 0)$.
- (iii) Coordinate planes divide the space into 8 octants.

Syed Humeera



Sn Divyanshi



Huzair Maqbo

THANK
You